Paleomagnetism of Franciscan Ultramafic Rocks from Red Mountain, California

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Results from the study of the magnetism in serpentinized ultramafic rocks of the Red Mountain intrusion southeast of San Francisco, California, suggest that these rocks possess a chemical remanent magnetization acquired during serpentinization. The stability of magnetization decreases with serpentinization, owing to the growth of larger magnetic grains. Thus, in highly serpentinized peridotites (serpentinites), most of the CRM is destroyed and the NRM becomes mainly viscous remanent magnetization, rendering these rocks unsuitable for paleomagnetic work. On the other hand, in partially serpentinized peridotites, pyroxenites, and dunites, the CRM is highly stable and paleomagnetically reliable. A study of the stable directions of magnetization indicates that the Red Mountain ultramafic body was emplaced and serpentinized prior to the folding of the Franciscan formation. The stability of magnetization and removal of viscous components were verified by the results of a storage test and alternating-field demagnetization experiments, by the divergence of the mean directions of magnetization from that of the present field, and, most important, by the convergence of the directions after tilt correction. Paleomagnetic pole positions calculated from the mean directions suggest that the magnetic pole rapidly migrated southward into the Atlantic Ocean during late Mesozoic, possibly more than once, as proposed previously by Grommé and Gluskoter.

INTRODUCTION

A study of ultramafic rocks covering different degrees of serpentinization was conducted for the purpose of investigating their magnetic properties. Samples were collected from Red Mountain, Mount Boardman Quadrangle, California, a large ultramafic body intruding the uppermost part of the exposed Franciscan section in the Diablo Range (Figure 1). The investigation consisted of two parts: (1) a study of intensity of induced and remanent magnetism, and (2) a study of directions of remanent magnetism.

Results of the first study, described in another paper [*Saad*, 1969], indicated that the susceptibility and intensity of remanent magnetism are highly variable, depending largely on the degree of serpentinization and, to a lesser extent, on the original rock composition. The remanent magnetization in these ultramafic rocks was found to be mainly a chemical remanent magnetization (CRM) acquired during the process of serpentinization whereby iron atoms, released from the silicate structure of the paramagnetic olivine and pyroxene, are oxidized to form ferrimagnetic magnetite. As serpentinization increases, the natural remanent magnetization increases in intensity but decreases in stability because of the growth of magnetite grains from single- to multi-domain size and of their oxidation, in some cases, to maghemite.

This paper deals with the paleomagnetic reliability and significance of the directions. The main objectives of the study were (1) to investigate more thoroughly the stability of the natural remanent magnetization (NRM) observed in these rocks and to confirm its origin, (2) to test whether ultramafic rocks can be used reliably for paleomagnetic work, and (3) to find the paleomagnetic pole position and examine its compatibility with previously reported poles from North America. The paleomagnetic data in this paper are analyzed in connection with the general problem of easterly directions found first by *Cox* [1957] and later by *Grommé and Gluskoter* [1965] from late Mesozoic and early Tertiary rocks of the western United States. The problem is of some interest since these

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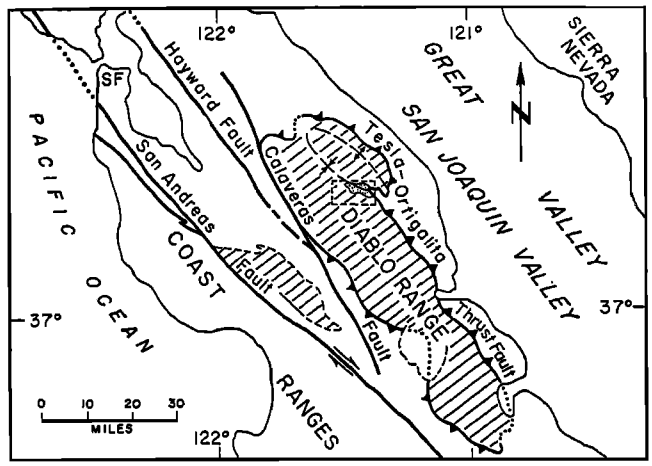


Fig. 1. Index map showing location of Red Mountain (small rectangle) in relation to the tectonic features of California Coast Ranges. Striped areas are Franciscan eugeosynclinal rocks of upper Jurassic to mid-Cretaceous.

anomalous directions may reflect real field excursions rather than tectonic rotations of the sampling sites.

Geologic setting. The geology of the area (Figure 2) is discussed in detail by *Maddock* [1955] and summarized elsewhere [*Saad,* 1968]. The Red Mountain intrusion is probably a sill or laccolith in the center of a large syncline roughly parallel to the strike of the intruded Franciscan sediments. The dip of the sedimentary beds near the contact is variable but generally greater than 60°. The axis of the Red Mountain syncline, which strikes N70°W on the average, is demarcated only approximately by a general reversal of dip over a broad area. The major folds in the area are not tightly compressed along their middle flanks and have average dips of around 40° [*Maddock,* 1964]. The intrusion is bounded by the Tesla-Ortigalita fault on the eastern side. On the other sides, the contact is believed to be a fault-contact also [*Bodenlos,* 1950]. Contact metamorphism along the margins is very slight, indicating that the body was intruded at fairly low temperature.

Serpentinized peridotite is the main rock type in the Red Mountain. The degree of serpentinization ranges from 20 to 95% and averages 65%. Dunite, which is 85% serpentinized on the average, occurs as small bodies, the largest of which forms a core in the eastern part of the intrusion (Figure 2). Pyroxenite occurs locally as small pods and veins and is about 20% serpentinized. Serpentinite, in this study, refers to a 100% serpentinized rock with no remanent grains of olivine or pyroxene. Most of the samples belonging to this rock type were collected from a small sill south of the main body. Outlines of some pyroxene grains were observed in thin sections, suggesting that these rocks were originally peridotites rather than dunites.

The slight deformation of fabrics in partially serpentinized peridotites and dunites suggests that serpentinization occurred after final emplacement of the ultramafic rocks.

Age considerations. Most of the ultramafic bodies in the California Coast Ranges were emplaced either during or subsequent to Franciscan time but prior to upper Cretaceous [*Bailey et al.,* 1964; *Burch,* 1968; *Hawkes et al.,* 1942]. *Maddock* [1964] deduced that the Franciscan rocks in the Diablo Range are pre-uppermost Jurassic age. The Red Mountain ultramafic rocks are intrusive into the uppermost part of the Franciscan section, indicating that they are upper Jurassic to upper Cretaceous in age and more probably upper Jurassic. This conclusion is supported by the fact that no ultramafic bodies intrude the adjacent upper Cretaceous sediments.

The upper Cretaceous can be chosen also as an upper limit for the age of folding since the Red Mountain syncline is truncated by the

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Tesla-Ortigalita fault and apparently does not extend into the upper Cretaceous sediments. Serpentinization occurred after final emplacement of the ultramafic rocks, as mentioned above, and prior to folding as indicated by the directions of magnetization discussed later.

Field and laboratory procedures. A total of 17 sites were chosen for obtaining oriented samples along road cuts and trails in the Red Mountain area (Figure 2). At several localities within each site, one to five samples were obtained by means of a portable gasoline-powered, water-cooled diamond core drill. The cores were oriented in situ with an accuracy of ±2° (s.d.) using a technique described by *Doell and Cox* [1967*a*]. Within each site, localities are separated by 30/100 meters and within each locality samples were cored 8 cm/1 meter apart. Several unoriented samples were obtained also. The cores were 2.49 cm in diameter and were cut into one to three specimens each (about 2.28 cm long) for remanent magnetization measurements, giving a total of 177 oriented specimens and 54 unoriented ones.

The intensity and direction of remanent magnetization were measured on a spinner magnetometer of the type described by *Doell and Cox* [1967*b*]. A digital computer was used to reduce the magnetometer readings to direction and intensity values with estimated accuracies of ±1.3° (s.d.) and ±6% (s.d.) respectively.

The apparatus used for demagnetization experiments is a three-axis tumbler that rotates the specimen in an alternating field smoothly reduced to zero from a peak value [*Doell and Cox*, 1967*c*]. The purposes of these experiments were to eliminate secondary soft components of viscous and isothermal remanent magnetization [*Cox*, 1961] and to establish the stability of magnetization. In order to reduce any anomalous magnetization added by the demagnetizing apparatus, most of the specimens demagnetized at fields of 200 oe or more were inverted in the tumbler and redemagnetized at the same field. The duplicate measurements were averaged and given unit weight in the analysis.

Polished sections and thermomagnetic measurements were used to identify the magnetic minerals in 20 specimens. The apparatus used for thermomagnetic measurements is an automatically recording quartz torsion balance described by *Clark* [1967]. The accuracy of the

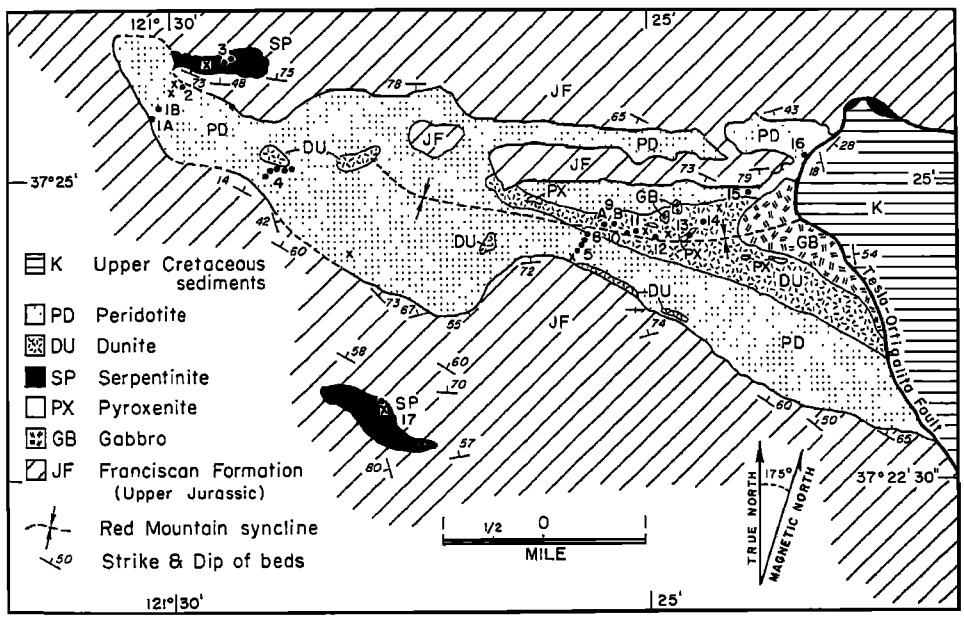


Fig. 2. Generalized geologic map of the Red Mountain area showing locations of collecting sites. Numbers associated with solid circles refer to sites of oriented samples. Crosses are site locations of unoriented samples. (Geology taken from *Maddock* [1964], *Bodenlos* [1950], and *Hawkes et al.* [1942]).

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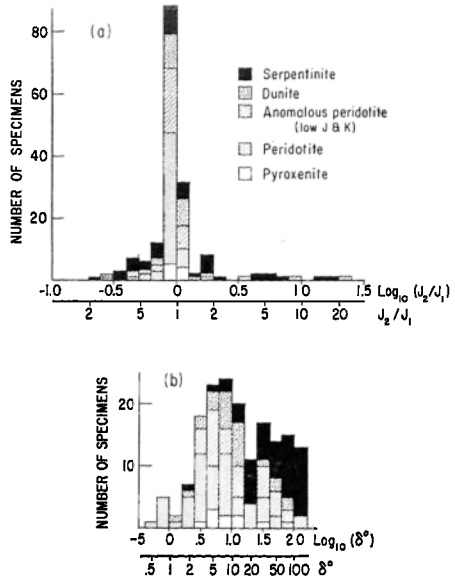


Fig. 3. Storage test histograms showing changes in (*a*) intensity of NRM, and (*b*) direction of NRM, with repeat measurements. J_1 and J_2, first and second measurements; $\delta°$, angular deviation between repeat measurements. Average time interval between measurements, 1.5 years.

Curie temperatures is estimated as ±5°C. Magnetic separates from one sample were analyzed by X-ray diffraction method.

Directions of magnetization were plotted on Lambert equal-area projections. All the data were treated statistically using the methods described by *Fisher* [1953] and *Wilson* [1959].

Results of Magnetic Measurements

Intensity of Natural Remanent Magnetization

The intensity of NRM (J) measured for all specimens varies from 2.7×10^{-5} to 2.9×10^{-3} emu/cc. The average intensities for the four rock types in emu/cc are: 1.79×10^{-4} for pyroxenite, 2.45×10^{-4} for peridotite, 2.26×10^{-4} for dunite, and 13.5×10^{-4} for serpentinite. The average values of Koenigsberger ratio (Q) are: 1.49 for pyroxenite, 1.40 for peridotite, 0.83 for dunite, and 0.59 for serpentinite. Some peridotite samples, referred to as 'anomalous' peridotites, have relatively low values of intensity of NRM (J) and susceptibility (K). The possible causes of their anomalous magnetization are discussed in another paper [*Saad*, 1969].

Polished sections, X-ray diffraction, and thermomagnetic measurements showed that magnetite, with Curie temperature of 550° to 580°C, is the main magnetic mineral in these ultramafic rocks.

Directions of NRM

The directions of the natural remanent magnetization of all specimens from the Red Mountain exhibit considerable initial scatter as a whole and within each site. The largest scatter exists within the serpentinite sites (e.g. SP03, Figure 6*a*). The most important causes of scatter are the acquisition of viscous magnetization, weathering, and tectonic movements of rocks after being magnetized. The following experiments were done to determine which of these processes had occurred.

Stability and Reliability Tests

Storage test. The initial magnetization of all specimens was measured twice in an average period of 1.5 years. Any significant change in magnetization indicates the existence of soft magnetic components that can be realigned by the earth's field. The results of the storage test are summarized in Figure 3.

Most of the serpentinites are unstable, with values of angular deviation δ up to 120° and with J changing by a factor of up to 20. The pyroxenites and most of the peridotites and dunites are stable with δ values of 1 to 15° and minor changes in intensity, indicating a considerable stability of NRM on the laboratory time scale. The minor changes in J are probably due in large part to magnetometer drift. Figure 3 also shows that the 'anomalous' peridotites have varying stabilities.

In almost all cases, a change in direction was accompanied by a comparable change in intensity between repeat measurements; on the average, the directional changes in degrees were approximately equal to the intensity changes in percent.

Alternating field demagnetization. A total of 35 specimens were subjected to progressive step demagnetization in peak alternating fields of 12.5, 25, 50, 100, 200, 400, and 600 oe [*As and Zijderveld*, 1958]. Figure 4 shows some examples of typical demagnetization curves for the various rock types and degrees of serpentinization. The curves are divided into three types.

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Type 1 is characteristic of stable magnetization whereby more than 50% of NRM remains after demagnetization in the 200-oe alternating field. This indicates that most of the magnetization resides in domains with high coercive forces. Curves of this type are typical of CRM or TRM [*Kobayashi*, 1959]. Type 2 curves exhibit moderate stability with 10 to 50% of NRM remaining after demagnetization in the 200-oe field, but type 3, with less than 10% of the NRM remaining at 200 oe, shows high instability. The pyroxenites, in general, are highly stable (type 1). The peridotites and dunites have curves of type 1 and 2, whereas most of the serpentinites exhibit maximum instability (type 3). The 'anomalous' peridotites, on the other hand, have various degrees of stability (Figure 5); although some are highly unstable, others are considerably stable (e.g. specimen PD09A1 in Figure 4). In general, the stability of NRM decreases with density. It was found that, in almost all cases, the samples with unstable ac demagnetization characteristics also change magnetically during storage. These experiments indicate that a varying portion of

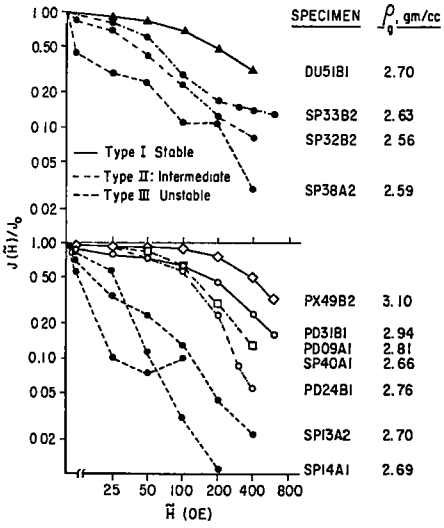


Fig. 4. Typical ac demagnetization curves showing stability of NRM for the different ultramafic rock types and various degrees of serpentinization. Ordinate is the fraction of original intensity of NRM remaining after demagnetization in alternating field of peak value $\tilde{H}$ oe. ρ_g, grain density; PX, pyroxenite; PD, peridotite; DU, dunite; SP, serpentinite.

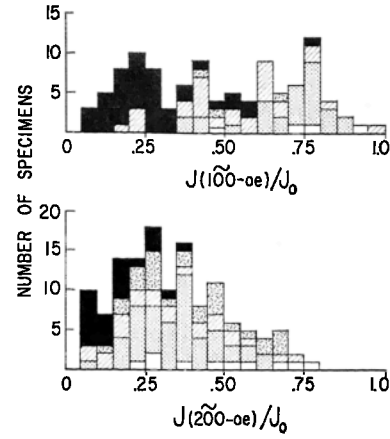


Fig. 5. Histograms of the fraction of original intensity of NRM remaining after demagnetization in 100- and 200-oe alternating fields. (See legend for Figure 3).

NRM is of viscous temporary origin and resides in the lower part of the coercive force spectrum.

To remove the unstable temporary components, all the specimens were subjected to partial demagnetization in an alternating field determined by using the criterion of minimum scatter for each site [*Irving et al.*, 1961], as illustrated in Figure 6. A field of 100 to 200 oe was sufficient to remove the temporary components, as suggested by the reduction in the scatter within all sites after demagnetization. Figures 6 and 7 show some typical examples of changes during alternating field demagnetization.

Consistency among directions. Although the directions of magnetization within each site are generally well grouped after partial demagnetization, a few specimens have divergent directions (Figure 7*b*). Of these, five from site PDO5 have a stable magnetization, whereas the rest are very unstable. The divergence of the directions in the case of stable magnetization is due to weathering. The directions of magnetization are displaced toward the present field direction, the highly weathered specimens showing maximum displacement. During surface weathering, new magnetic minerals are generated or pre-existing minerals are oxidized. Thus the rock acquires a new secondary component of chemical origin (CRM) that can be very stable and is in the direction of the

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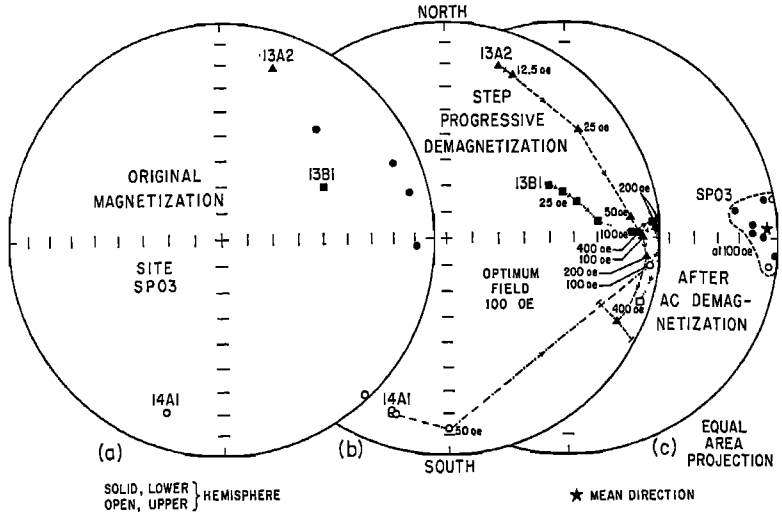


Fig. 6. Changes in direction of NRM during alternating field demagnetization. (a) Initial directions of site SP03. (b) Selection of an optimum demagnetizing field for site SP03; progressive demagnetization of three specimens shows that the scatter in their original directions is minimum at 100 oe. (c) Directions of site SP03 after ac partial demagnetization at 100 oe; the reduction in initial scatter indicates the removal of viscous components of magnetization.

present field [*Robertson and Hastie,* 1962]. The unstable specimens, however, have divergent directions due to the acquisition of components of anhysteretic remanent magnetization (ARM) during the demagnetization process. This is indicated by the repeat-demagnetization experiments explained earlier and by the increase in intensity of magnetization and the wandering in directions at higher fields (e.g. SP40A1, Figure 4).

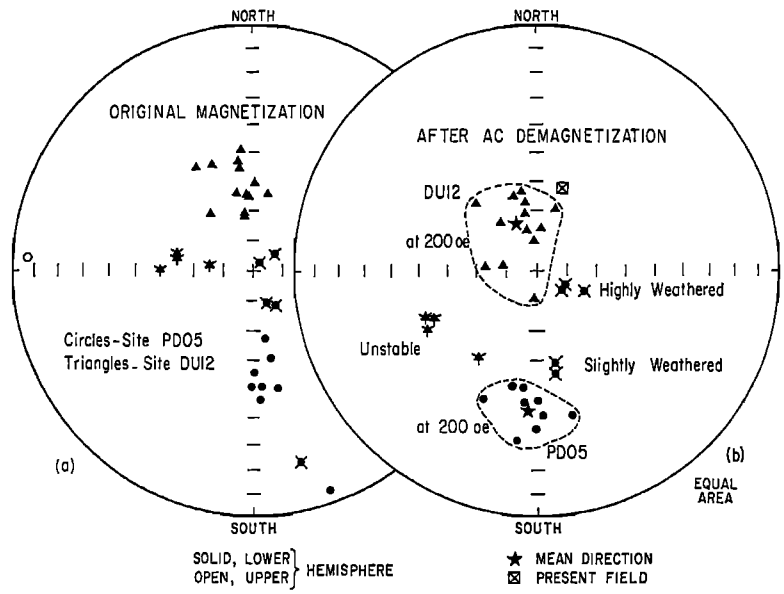


Fig. 7. Typical examples of grouping of directions after ac partial demagnetization and rejection of unstable (crossed triangles) and weathered (crossed circles) specimens.

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All specimens within each site, excluding those with divergent directions due to any of the above criteria, show consistency among their directions. Such internal consistency, together with the divergence of the mean directions of each site from that of the present earth's field, are indications of stability and nonviscous origin of magnetization.

Tilt test. Although the previous criteria assisted in testing the stability of the directions and in reducing their scatter within each site, the mean directions of the sites are significantly scattered as shown in Figure 9 (open circles). However, the rocks from which the samples were obtained are considerably tilted and folded. A plot of these directions along a profile perpendicular to the synclinal axis of the Red Mountain shows a pronounced systematic change in the mean directions across the structure (Figure 8), thus suggesting the necessity of applying Graham's tilt test [*Graham*, 1949]. The test has great significance in providing valuable information regarding the time of the acquisition of magnetization, its stability and possibly its origin.

The average strike and dip of the rocks at each site were determined from the regional geology of the area by interpolation from zero dip over the axis of the syncline to the strike and dip of the neighboring sediments at the margin of the concordant intrusion. Using the strikes and dips estimated by this procedure and given in Table 1, Graham's tilt test was applied to the mean directions of all sites. The results of the test are shown in Figure 9. With the exception of five sites (Figure 9*b*), the scatter of the mean directions is reduced markedly on

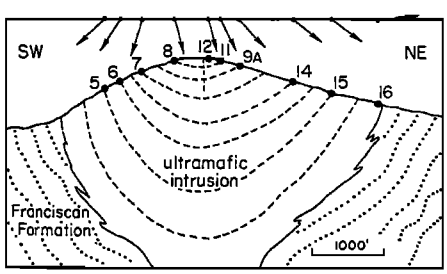


Fig. 8. Change in mean directions of NRM across the Red Mountain syncline. Structure is inferred. Arrows indicate directions of magnetization projected onto a plane perpendicular to the structural axis. Numbers refer to sites (Figure 2).

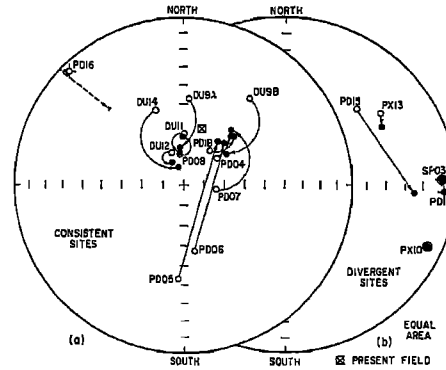


Fig. 9. Graham's tilt test applied to mean site directions of magnetization. Open circles, before test; solid circles, after test; all points except crossed ones, lower hemisphere. Tilt of site PD16, uncertain.

correcting for the inferred dip and strike. This indicates that the magnetization of these rocks predates the folding and has not changed since then, thus providing substantial evidence for the high stability of the magnetization. It is evident that these results exclude any post-folding thermoremanent magnetization (TRM) as an origin of the magnetization. A pre-folding TRM can be acquired under two circumstances:

1. During the cooling of the original peridotite magma below the surface of the crust, the ultramafic body acquired a stable but weak TRM due to the scarcity of magnetic minerals in the fresh rocks. However, the directions of magnetization would be expected to be scattered (after the tilt test) due to tectonic emplacement.

2. The intrusion was strongly magnetized due to serpentinization but then was reheated to its Curie temperature and acquired a TRM on later cooling. This is unlikely since reheating of the rocks above 500°C causes deserpentinization. Moreover, a second episode of sepentinization will cause scattering in the directions, in contradiction to the results of the tilt test.

In summary, the results of the tilt test exclude TRM as the origin of magnetization. Thus, the remanent magnetization of the Red Mountain ultramafics is probably CRM developed during the serpentinization of the body prior to its folding. Except for the pyroxenites, the effect of the initial TRM is negligible either

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TABLE 1. Statistical Analyses of Mean Site Directions of Magnetization for Red Mountain

Site	Strike, deg	Dip, deg	Field, oe	N	R	Before DR	Before IR	After DR	After IR	k	$A95$	ASD	SDM
PD1A	N35W	14NE	200	4	3.965	91.8	4.6	92.0	−8.0	86.2	10.0	7.6	3.8
PD1B	N35W	14NE	100	12	11.643	42.8	67.6	45.0	56.0	30.8	7.9	14.0	4.0
PD02	rejected		100	4	2.709	220.5	29.9	···	···	2.3	79.5	53.2	26.6
SP03	N75W	60NE	100	5	4.936	87.8	9.8	87.5	−9.5	62.1	9.8	9.2	4.1
PD04	N58W	14NE	100	16	15.339	50.2	68.5	44.0	56.5	22.7	7.9	16.5	4.1
PD05	N74W	72NE	200	5	4.940	182.4	42.9	38.6	62.0	66.8	9.4	8.9	4.0
PD06	N75W	55NE	200	2	1.998	171.6	56.9	45.0	61.5	413.5	12.3	2.8	2.0
PD07	N77W	30NE	200	2	1.989	100.6	75.3	41.1	55.0	89.1	26.8	6.1	4.3
PD08	N80W	10NE	200	2	1.996	354.6	75.7	359.0	65.8	241.9	16.1	3.7	2.6
DU9A	N80W	25SW	200	1	···	3.8	47.4	353.1	71.8	···	···	···	···
DU9B	N70W	32SW	200	4	3.969	37.1	34.9	54.8	64.2	96.1	9.4	8.3	4.1
PX10		0	200	2	1.987	113.3	9.9	113.3	9.9	76.5	29.0	6.6	4.6
DU11	N75W	10SW	100	2	1.985	1.2	65.4	354.1	74.5	65.2	31.5	7.1	5.0
DU12	N87W	5SW	200	9	8.794	334.6	73.4	329.9	78.0	38.8	8.4	12.3	4.1
PX13	N80E	10SE	200	5	4.911	52.4	29.1	58.0	33.8	44.8	11.5	10.8	4.8
DU14	N70E	30SE	200	2	1.998	339.0	50.3	339.0	81.5	483.7	11.4	2.6	1.8
PD15	N60E	70SE	200	1	···	42.3	37.4	93.2	25.5	···	···	···	···
PD16	N40E	80SE	200	2	1.999	315.1	−6.2	329.5	74.0	968.2	8.0	1.8	1.3
SP17	rejected		0	30	21.717	23.1	86.9	···	···	3.5	16.6	43.3	7.9

Notes.

N Number of samples (number of specimens for rejected sites).
R Vector sum of N unit vectors.
DR Declination of the mean vector.
IR Inclination of the mean vector.
k Fisher precision parameter $= (N - 1)/(N - R)$.
$A95$ Radius of 95% confidence cone about direction of R.
ASD Angular standard deviation [*Wilson*, 1959].
SDM Standard deviation of the mean [*Wilson* 1959].
Before and After refer to tilt test.
Field refers to peak value of demagnetizing field in oe.

because it is weak compared with the strong CRM, or possibly because it was destroyed during serpentinization. The pyroxenites, being slightly serpentinized, may have an initial TRM stronger than the later CRM.

Paleomagnetic Analysis

Analysis of average directions. The average directions of magnetization and precision parameters for each site were computed using the methods described by *Fisher* [1953] and *Wilson* [1959]. The statistical methods were applied twice on two levels. On the first level, sample means were obtained by using the methods on specimen observations from a sample giving each unit weight. On the second level, each sample mean from a site was given unit weight to obtain site means. This procedure was used because specimen-to-specimen variations are generally smaller than variations among samples although in some cases the first are as large as the latter [*Irving*, 1964, chap. 4]. Other procedures have been tried, e.g. giving unit weights to each specimen to find site means. In all cases, the differences between the site means obtained were not more than 2°. Table 1 shows the statistics on the mean site directions obtained by giving a unit weight to each sample mean.

The mean site directions, in general, are found to fall into three classes. The first class includes two sites (PD02 and SP17) with widely scattered directions, as indicated by a very low precision parameter k. The cause is uncertain for site PD02, although large errors in measuring the small moments of these samples and large amounts of ARM acquired during demagnetization may be responsible. Site SP17 includes all serpentinite specimens collected from the small sill south of the main body (Figure 2). These

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specimens are characterized by unstable intensities, scattered directions, and very rapid changes in both direction and intensity of NRM while being measured in the spinner magnetometer; the changes are more pronounced as the rock becomes more demagnetized. The phenomenon is due to a very soft component of magnetization, with short relaxation time of the order of minutes or seconds superimposed on other components in the rock. Another feature of the serpentinites is the pronounced wandering of directions from one hemisphere to the other during the process of demagnetization at higher fields.

The second class of results is characterized by magnetization possessing stability and consistency within each site yet having a mean direction that is inconsistent with other site means after applying a tilt correction (Figure 9*b*). The divergence of these means can be explained by the occurrence of one or both of the following processes: (1) the magnetization was acquired at a time different from that of the main body, or (2) the divergent sites were rotated after being magnetized. These sites include two pyroxenite sites (PX10, a vein, and PX13, a pod) possibly intruded later into the main ultramafic body at different times. Three other sites that belong to this class are serpentinites and serpentinized peridotites. They are near the fault contact of the body and may have rotated. However, in spite of the difference in rock types and the wide geographic separation between them (Figure 2), all have about the same anomalous direction, which is possibly a reflection of an anomalous paleomagnetic field direction.

All other sites (third class) have stable magnetic directions that are mutually consistent after the tilt correction (Figure 9*a*). The mean directions, however, form two slightly separated groups, one mostly dunite and the other peridotite. The F-ratio test was used to compare the precision parameters of the two groups and also to compare their mean directions [*Watson and Irving*, 1957]. The test showed that the two precision parameters are identical and that the separation of the two mean directions is significant. This separation supports the view of *Hawkes et al.* [1942] that the dunite was intruded at a time subsequent to the intrusion of the peridotite and may reflect also a change in the paleomagnetic field direction.

Paleomagnetic pole positions and comparison with other data. To calculate the paleomagnetic poles for the purpose of comparison with other data, the sites are treated statistically as three independent groups: the peridotites, the dunites and the divergent sites. The results of the statistical analysis shown in Table 2 were obtained by giving each mean site direction a unit weight. Paleomagnetic poles corresponding to each group-mean direction are given in Table 3, together with three late Mesozoic poles reported from California [*Grommé et al.*, 1967].

It is evident that the Red Mountain paleomagnetic poles, particularly poles (1) and (3), have anomalous easterly positions compared to all other published Jurassic and Cretaceous poles for North America that have longitudes between 90° and 270°W [*Irving*, 1964]. The only comparable pole for this time with easterly longitude has been obtained by *Grommé and Gluskoter* [1965] from some diabasic and spilitic

TABLE 2. Statistical Analyses of Group-Average Directions of Magnetization for Red Mountain

Group	I	D	N	R	k	$A95$	ASD	SDM
(1)	59.3	44.5	6	5.983	302	3.9	4.7	1.9
(2)	74.6	350.0	5	4.972	144	6.4	6.7	3.0
(3)	10.8	89.9	5	4.566	9	26.6	26.7	11.9
(1) + (2)	68.2	28.8	11	10.702	34	8.0	14.0	4.2

Notes.
(1) Peridotite sites.
(2) Dunite sites.
(3) Other divergent sites (including pyroxenite sites).
N Number of sites.
Other symbols as in Table 1.

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TABLE 3. Paleomagnetic Pole Positions

Location	Pole	N Lat.	W Long.	*DM*	*DP*	Age (10^6 Years)
Red Mountain ultramafics	(1)	55.6	49.9	5.8	4.3	Upper Jurassic to Upper Cretaceous (80–140, estimated)
	(2)	65.6	132.0	11.6	10.5	
	(3)	3.4	35.8	27.0	13.7	
	(1) + (2)	65.5	75.1	13.4	11.3	
Franciscan formation	*F*	26	50	17.8	11.5	
Sierra Nevada plutons	*J*	53.7	174.1	9.5	9.5	138
	C	69	165	9.6	9.6	84

Notes.

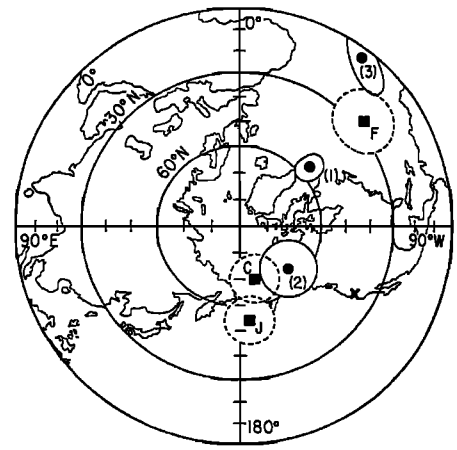
(1), (2), (3) Poles calculated in the present study from information in Table 2.

F, J, C Upper Jurassic and Cretaceous poles from California (for comparison). *F*: *Grommé and Gluskoter* [1965]; *J, C*: *Grommé et al.* [1967].

DM, DP Semiaxes of 95% oval of confidence.

bodies in the Franciscan formation (late Jurassic—late Cretaceous) in California (pole *F*, Figure 10). Pole (3) from the divergent sites of the Red Mountain is located close to the anomalous Franciscan pole; poles (1) and (2) from the peridotite and dunite groups respectively are located between this anomalous position and the currently accepted positions of the Jurassic and Cretaceous poles for western U.S.A. (poles *J* and *C*, Figure 10). Pole (2), however, is very close to pole *C* from the Sierra Nevada granites of Cretaceous age [*Grommé et al.*, 1967]; the overlap of their ovals of confidence indicates that they are not significantly different at the 95% probability level.

There are two other poles that have been reported with anomalous easterly directions. The first is from lower Eocene Siletz River volcanics of western Oregon [*Cox and Doell*, 1960] and has a pole position of 37°N, 49°W, similar to the Franciscan pole but with 10° higher latitude. The other pole is from late Mesozoic dolerites from Spitsbergen [*Spall*, 1968] at 60°N, 2°E. The question arises as to whether these easterly directions of the magnetic field found from Siletz River volcanics, Franciscan formation, Spitsbergen dolerites, and Red Mountain ultramafics are due to tectonic rotation of the sampling sites or real excursions of the geomagnetic field. The paleomagnetic data from the Red Mountain suggest that the geomagnetic pole, assuming a mainly dipolar field, migrated southeastward into the Atlantic Ocean and back on one or more occasions during late Mesozoic and probably early Eocene as proposed previously by *Grommé and Gluskoter* [1965]. In every case mentioned above, there is no evidence other than the magnetic data in support of tectonic rotations. Furthermore, the fact that similar data have been ob-



EQUAL-AREA PROJECTION OF THE NORTHERN HEMISPHERE

Fig. 10. Paleomagnetic poles from the Red Mountain ultramafic body (circles) and other late Mesozoic poles (squares). (1), (2), and (3), poles from peridotite, dunite, and pyroxenite and marginal sites, respectively (present study); *F*, pole from Franciscan formation [*Grommé and Gluskoter*, 1965]; *J* and *C*, poles from Sierra Nevada plutons of upper Jurassic and Cretaceous, respectively [*Grommé et al.*, 1967]. 95% confidence ovals or circles are shown solid for present study and dashed for others. X indicates sampling location.

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tained from Spitsbergen in another continent indicates a worldwide phenomenon related to the geomagnetic field itself and not to local tectonic rotations.

Although accurate dating of the Franciscan formation, including the ultramafic rocks, is necessary for evaluating any paleomagnetic data, some conclusions regarding the emplacement and magnetization of the Red Mountain body can be summarized in light of the above mentioned hypothesis as follows:

1. The margins of the body were magnetized first (by serpentinization) during or immediately after emplacement at a time closer to Franciscan magnetization time when the magnetic pole was in one of its excursions into the Atlantic Ocean (pole 3, Figure 10).
2. The main body (peridotites) acquired its magnetization at a later time during which the magnetic pole was recessing toward its normal position (pole 1, Figure 10).
3. The dunites were emplaced and magnetized at a much later time when the magnetic pole was near its normal position, possibly in the Cretaceous (pole 2, Figure 10).
4. The pyroxenites, which intruded the dunites as veins and pods, were magnetized after the dunites, possibly in the Eocene time during which the magnetic pole was in another excursion as suggested by *Grommé and Gluskoter* [1965] (pole 3, Figure 10). It is also possible that the anomalous directions of the pyroxenites are due to the stronger effect of the initial TRM on the later CRM.

Conclusions

Results of the tilt test, applied to the average directions of magnetization, support the conclusion of the intensity study that the remanent magnetization observed in the Red Mountain ultramafic rocks is CRM acquired during the serpentinization of the body. The results also showed that the body was emplaced and serpentinized prior to folding, a finding of importance to the geologic history of the Coast Ranges but difficult to obtain geologically [*Bailey et al.*, 1964, p. 83].

In using ultramafic rocks for paleomagnetic work, only the partially serpentinized are reliable; the highly serpentinized are magnetically unstable. Paleomagnetic pole positions, found from the study of partially serpentinized ultramafics of the Red Mountain, indicate relatively rapid changes in the geomagnetic field during the late Mesozoic, thus confirming the hypothesis advanced by *Grommé and Gluskoter* [1965].

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References

As, J. A., and J. D. A. Zijderveld, Magnetic cleaning of rocks in paleomagnetic research, *Geophys. J., 1,* 308–319, 1958.

Bailey, E. H., W. P. Irwin, and D. L. Jones, Franciscan and related rocks and their significance in the geology of western California, *Calif. Div. Mines Geol. Bull. 183,* 177 pp., 1964.

Bodenlos, A. J., Geology of the Red Mountain magnesite district, Santa Clara and Stanislaus Counties, California, *Calif. J. Mines Geol., 46,* 223–278, 1950.

Burch, S. H., Tectonic emplacement of the Burro Mountain ultramafic body, Santa Lucia Range, California, *Bull. Geol. Soc. Amer., 79*(5), 527–544, 1968.

Clark, H. C., A fused-quartz Curie-point balance, in *Methods in Paleomagnetism,* edited by K. M. Creer, D. W. Collinson, and S. K. Runcorn, pp. 438–439, Elsevier, Amsterdam, 1967.

Cox, A., Remanent magnetization of lower to middle Eocene basalt flows from Oregon, *Nature, 179,* 685–686, 1957.

Cox, A., Anomalous remanent magnetization of basalt, *U.S. Geol. Surv. Bull., 1083-E,* 131–160, 1961.

Cox, A., and R. R. Doell, Review of paleomagnetism, *Bull. Geol. Soc. Amer., 71,* 645–768, 1960

Doell, R. R., and A. Cox, Paleomagnetic sampling with a portable coring drill, in *Methods in Paleomagnetism,* edited by K. M. Creer, D. W. Collinson, and S. K. Runcorn, pp. 21–25, Elsevier, Amsterdam, 1967*a*.

Doell, R. R., and A. Cox, A spinner magnetometer for igneous rocks, in *Methods in Paleomagnetism,* edited by K. M. Creer, D. W. Collinson, and S. K. Runcorn, pp. 136–141, Elsevier, Amsterdam, 1967*b*.

Doell, R. R., and A. Cox, Analysis of alternating field demagnetization equipment, in *Methods in Paleomagnetism,* edited by K. M. Creer, D. W. Collinson, and S. K. Runcorn, pp. 241–253, Elsevier, Amsterdam, 1967*c*.

21562202, 1969, 27, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB074i027p06567 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Fisher, R. A., Dispersion on a sphere, *Proc. Roy. Soc. London, A, 217,* 195–305, 1953.

Graham, J. W., The stability and significance of magnetism in sedimentary rocks, *J. Geophys. Res., 54,* 131–167, 1949.

Grommé, C. S., and H. J. Gluskoter, Remanent magnetization of spilite and diabase in the Franciscan formation, western Marin County, California, *J. Geol., 73,* 74–94, 1965.

Grommé, C. S., R. T. Merrill, and J. Verhoogen, Paleomagnetism of Jurassic and Cretaceous plutonic rocks in the Sierra Nevada, California, and its significance for polar wandering and continental drift, *J. Geophys. Res., 72,* 5661–5684, 1967.

Hawkes, H. E., Jr., F. G. Wells, and D. P. Wheeler, Jr., Chromite and quicksilver deposits of the Del Puerto area, Stanislaus County, California, *U. S. Geol. Surv. Bull., 936-D,* 79–110, 1942.

Irving, E., *Paleomagnetism and its Application to Geological and Geophysical Problems,* 399 pp., John Wiley and Sons, New York, 1964.

Irving, E., P. M. Stott, and M. A. Ward, Demagnetization of igneous rocks by alternating magnetic fields, *Phil. Mag., 6,* 225–241, 1961.

Kobayashi, K., Chemical remanent magnetization of ferromagnetic minerals and its application to rock magnetism, *J. Geomagnet. Geoelec., Kyoto, 10,* 99–117, 1959.

Maddock, M. E., Geology of the Mount Boardman Quadrangle, California, Ph.D. thesis, University of California, Berkeley, 1955.

Maddock, M. E., Geology of the Mount Boardman Quadrangle, Santa Clara and Stanislaus Counties, California, *Calif. Div. Mines Geol. Map Sheet 3,* 1964.

Robertson, W. A., and L. Hastie, A paleomagnetic study of the Cygnet alkaline complex of Tasmania, *J. Geol. Soc. Australia, 8,* 259–268, 1962.

Saad, A. H., Magnetic properties of ultramafic rocks from Red Mountain, California, Ph.D. thesis, Stanford University, Stanford, California, 1938.

Saad, A. H., Magnetic properties of ultramafic rocks from Red Mountain, California, *Geophysics,* in press, 1969.

Spall, H., Anomalous paleomagnetic poles from late Mesozoic dolerites from Spitsbergen, *Earth Planet. Sci. Lett., 4,* 73–78, 1968.

Wilson, R. L., Remanent magnetism of late Secondary and early Tertiary British rocks, *Phil. Mag., 4,* 750–755, 1959.

Watson, G. S., and E. Irving, Statistical methods in rock magnetism, *Mon. Not. Roy. Astron. Soc., Geophys. Suppl., 7,* 289–300, 1957.

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